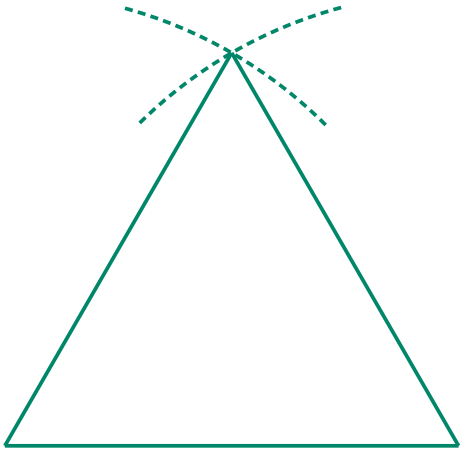
GEOMETRY QUEST · WORLD 13

Constructions

ANSWER SHEET · for the grown-up

Answers are shown in green. “Check” means look at the child’s drawing or measuring: allow ±2° or ±2 mm.

1 Construct an equilateral triangle with sides of 6 cm. Only measure the first side; use your compass for the rest. Leave your arcs showing!



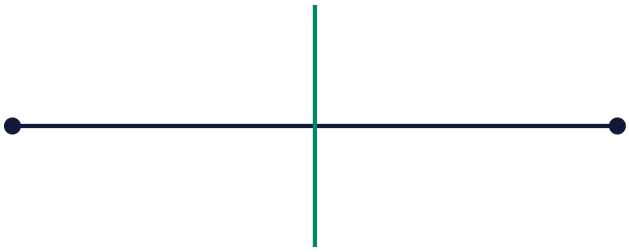
1. Draw AB = 6 cm.
2. Compass on A, open to B. Draw an arc.
3. Same opening, compass on B. Draw an arc.
4. Where the arcs cross is C. Join AC and BC.

Check: all sides 6 cm, all angles 60°.

2 Can these three sticks make a triangle? Write yes, no, or flat.

3, 4, 5 cm	2, 2, 5 cm	5, 5, 10 cm	6, 7, 8 cm
yes	no	flat	yes

3 Bisect (cut exactly in half) this 8 cm line with a compass: open it more than halfway, draw arcs above and below from each end, and join the crossings.



Check: 4 cm each side

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4 Construct a triangle with sides 5 cm, 6 cm and 7 cm. (Draw the 7 cm side, then use arcs of 5 cm and 6 cm from its ends.)

Check the three sides with a ruler: 5, 6 and 7 cm.

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